Raghavan 2021 PDAC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (sample, θ=5): same-sample neighbour fraction 0.94→0.26 (plateau by θ=2); samples intermix per-neighbourhood (iLISI) on one continuous connected manifold; 0 detached islands removed.

Gate 2 — module geography → GO (non-branching)

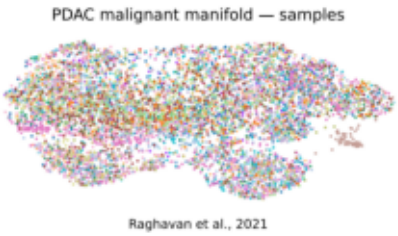
Non-branching: Gast & pSom co-vary (smoothed r=+0.32, joint-high enrichment 1.15 = overlap) and both oppose Crest (r=−0.80 / −0.69); single Gast/pSom→Crest gradient with Crest terminal (decile pseudotime 0.68). Classical/basal plasticity axis is real (r=−0.70) but ORTHOGONAL to the embryonic modules (pSom vs classical/basal +0.04/+0.05) — not the developmental bifurcation.

Gate 3 — cross-repression → NA

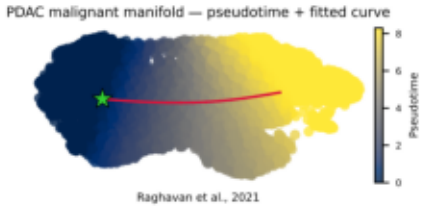
No Gast/pSom bifurcation (they overlap rather than cross-repress) → no hub to test. Gate 3 not applicable.

STEP FIGURES

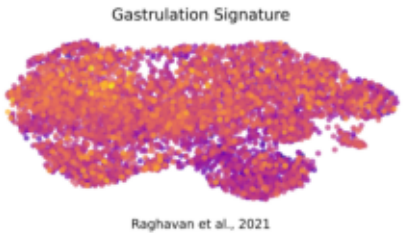
Manifold — samples



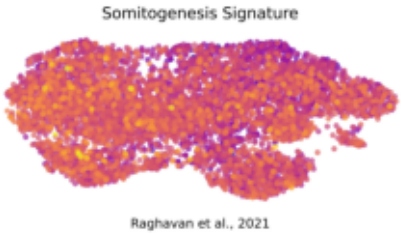
Pseudotime



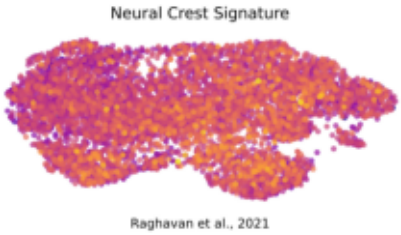
Gastrulation



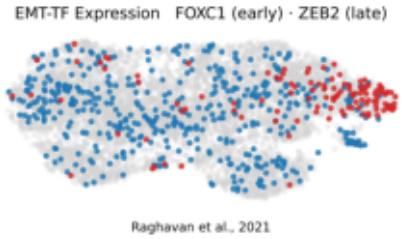
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic ductal adenocarcinoma (Raghavan et al., 2021), Seq-Well — 7,168 malignant cells (20 samples); disease-restricted to PDAC (662 PANET cells excluded). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching): a single Gast/pSom→Crest developmental axis with Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. The classical/basal plasticity axis is real (r=−0.70) but orthogonal to the embryonic modules — not the developmental bifurcation. EMT-TFs FOXC1 (early/root) → ZEB2 (late/terminus).

Parameter choices

Sample threshold	≥100 cells; 20/22 samples kept (45 & 29-cell samples dropped)	**Disease restriction**	PDAC only (662 PANET cells, patient PANFR0580, excluded)
Overdispersed genes	1,308; PCA 50PC; Harmony(sample, θ=5)	**Embedding**	knn n_pcs=20, n_neighbors=30; UMAP spread=1.3
Trajectory	scFates curve N=15, λ=0.3, μ=100 (2 tips, 0 forks); corr(x,pt)=+0.97	**Root**	Gast/pSom-high tip
Geometry	Linear (non-branching); classical/basal axis orthogonal	**EMT-TFs**	FOXC1 (early/root) → ZEB2 (late/terminus); 3% floor, no VIM